

A-10. SALT-1 (Sloping Agricultural Land Technology)



Agriculture Land Conservation



Medium



Medium



Measure Type

Cost

Preferences by Community

Hazards

Implementation Time of Measures

Descriptions

Measures

Shifting cultivation and plough agriculture on slopping area accelerates land degradation, forest degradation, and biodiversity destruction in hilly region through deforestation and burning. As a result, the whole environment is disrupted. Among all, plough cultivation, pineapple and tobacco cultivation are posing a great threat to the agriculture, forests, and environment in hilly areas.

Bangladesh Forest Research Institute (BFRI) has developed Sloping agricultural land technology (SALT) in the hills for sustainable land management. In SALT, annual crops are grown between hedgerows in bands measuring 40-50 m wide. Sloping Agricultural Land Technology (SALT-1) is a simple, applicable, low-cost, but effective agroforestry technology with agricultural crops and forestry in a ratio of 75:25. Based on the experiences of MBRLC/ARLDF, this technology decreases erosion substantially compared to the traditional upland farming management practices and slows the flow of rainfall down the slope. In addition, it increases crop yield five to six-fold.

The sustainability of SALT as an agroforestry practice is associated with slow but continuous improvement of soil fertility status by proper management of the systems organic matter, decreasing erosion and slowing the flow of rainfall down the slope.

Location

R-114	R-99	R-65	K-98	K-76	K-71	B-173	B-77	B-76
■	■	■	■	■	■	■	■	■

Applicable Sites

- Erosion-Prone Agricultural Lands
- Shifting Cultivation (Jhum) Conversion Zones
- contour farming and agroforestry-based soil conservation.

Slope

Less than 3%	3% to 15%	15% to 30%	30% to 60%	Above 60%
		√	√	√

Design options and performance

- SALT-1 (Basic Agroforestry Model) – Crops (75%) planted between tree-based hedgerows (25%) to control erosion.
- SALT-2 (Livestock Integration Model) – Incorporates forage crops for sustainable agro-pastoral systems.
- SALT-3 (Soil Fertility Focused) – Uses nitrogen-fixing species (e.g., Gliricidia, Leucaena) for organic soil enrichment.
- SALT-4 (Agroforestry-Forest Protection Model) – Integrates perennial trees, timber species, and food crops to restore forest cover.

	- Multi-Strata Agroforestry Model – Combines annual crops, fruit trees, and medicinal plants for biodiversity conservation. Indigofera tasmanilas; Ipil-ipil; Desmodium spp.; Gliricidia - Crops: maize, sesame, chilli, papaya, etc.			
Feasibility criteria	Technical Feasibility - Ensures wide applicability in hill farming systems. - Reduces runoff and improves infiltration. - Uses simple, low-cost techniques with minimal training required. Environmental Feasibility - Reduces erosion and stabilizes land. - Slows rainfall movement, improving groundwater recharge. - Promotes native species cultivation alongside crops. Economic Feasibility - Utilizes existing farm inputs and minimal external investment. - Yields 5-6 times more than traditional upland farming. - Farmers benefit from timber, fruits, forage, and staple crops. Social Feasibility - Encourages farmer-led adoption and management. - Ensures sustainable agricultural production. - Aligns with Bangladesh's sustainable agriculture and afforestation programs.			
Operation and maintenance	- Regular Hedgerow Pruning & Soil Mulching – Maintains nutrient recycling and weed control. - Contour Planting & Erosion Monitoring – Prevents runoff and maintains structural integrity. - Intercropping & Crop Rotation – Enhances soil fertility and pest resistance. - Livestock Grazing Control – Protects vegetative barriers from overgrazing.			
Cost	Medium			
Benefits				
	Type	Low	Medium	High
	Canopy Cover Increase			
	Carbon Sequestration			
	Crop Production Increase			
	Crop Failure Risk Reduction			
	Ensuring Food Security			
	Fruit Production Increase			
	Forestry Goods Production Increase			
	Livestock Production Increase			
	Fisheries Production Increase			
	Alternative Livelihood Generation			
	Increase of Income			
	Soil Erosion Control			

	Improved Soil Health			
	Soil Fertility Increase			
	Sustainable Water Supply			
	Water Quality Improvement			
	Water Purification			
	River Bank Erosion Control			
	Surface Runoff Reduction			
	Groundwater Recharge			
	Pest and Disease Management			
	Reduction of Pesticides and Fertilizer Use			
	Ecosystem and Biodiversity Increase			
	Better Communications			
	Better Education			
	Better WASH facilities			
Implementation Challenges				
	Type	Low	Medium	High
	Land unavailable/high land area required			
	Supply chain disconnection			
	Land right related problems			
	High human investment capital need			
	Insufficient infrastructures			
	Less demand of agroforestry product			
	Environmentally unsustainable			
	Socially unacceptable			
	Less beneficial			
Environmental performance	- Reduced Soil Erosion & Degradation – Stabilizes fragile hill slopes. - Water Conservation & Rainfall Absorption – Enhances moisture retention and groundwater recharge. - Biodiversity Enhancement – Restores native tree cover and agroforestry landscapes. - Climate Resilience & Carbon Sequestration – Strengthens ecosystem adaptation and mitigates greenhouse gas emissions.			
Remarks	This measure will aid landscape restoration in CHT of Bangladesh along with creating alternative livelihood opportunities.			